

The Portability CIN: Calibrated Counterfactual Projection of NHL Player Impact under Environment Interventions

OptBot Research

Technical Report v1.0 — July 2026 — Confidential

Abstract

We present the Portability Causal Intervention Network (CIN), a system that projects a professional hockey player’s on-ice impact in an environment he has never played in — a new team, line, and role — before a single shift is observed there. The central design thesis is that credible causal AI is achieved not by architectural scale but by four enforceable commitments: (C1) answering only intervention classes that occur as natural experiments in the data, with tiered refusal outside support; (C2) structural deconfounding of player talent from deployment via a frozen, out-of-fold talent channel; (C3) optimizing the model on the distribution it will be queried on, by up-weighting windows adjacent to observed context switches; and (C4) conformal calibration of uncertainty on the intervention distribution itself, with achieved coverage attached to every emitted interval. On a frozen-clock backtest of 635 real player–team transitions (2021–2025), the minimal instantiation of the system (v0: destination-environment expectation plus a shrunken residual talent prior) attains RMSE 5.30 on first-40-game on-ice expected-goal share, versus 5.80 for a Marcel-style industry baseline (−8.6%; bootstrap 95% CI on the difference $[-0.67, -0.33]$, excluding zero) and 7.40 for last-season carryover. Emitted 80% intervals achieve 79.8% empirical coverage. We give the complete data substrate, estimators, neural architecture, leakage-audit methodology (including three data corruptions whose discovery materially changed the system), and an 18-item threat model.

1 Introduction

Player acquisition in the National Hockey League is a capital-allocation problem executed under a specific epistemic handicap: all available statistics describe a player’s output *in his previous environment*. On-ice results in hockey are violently contextual — linemates, opposition, deployment, coaching system, and schedule jointly determine a large fraction of observed variance — so the acquisition decision is, implicitly, a causal question: *what would this player’s output be under our environment?* Formally it is a counterfactual query, and the industry answers it today with heuristics (scouting intuition) or context-free extrapolations (weighted averages of past seasons).

This report documents a system that answers the query directly and, more importantly, *auditably*. Our position is that “causal AI” fails in practice for reasons that are disciplinary rather than architectural: systems answer interventions their data never contained, entangle treatment assignment with subject quality, extrapolate without support diagnostics, and report uncertainty that has never been checked against realized frequencies. Sections 2–8 present a design in which each of these failure modes is closed by an explicit, testable mechanism rather than by optimism.

Two results anchor the report. First, the minimal version of the system already clears the strongest public baseline with a confidence interval excluding zero (Section 9). Second, the engineering audit that preceded any modeling (Section 4) discovered three silent data corruptions — including an outcomes column that was identically zero across 761,823 rows — whose repair retroactively explains the failure of a previous uncertainty model. We consider the audit methodology as much a contribution as the architecture.

2 Problem Formulation

2.1 Units and notation

The atomic observational unit is the *window*: a maximal interval of game time with constant on-ice personnel and manpower state, indexed w . Each window carries, for a focal player p present in it:

- exposure $s_w > 0$ (the focal player’s seconds on ice in w);
- an environment description $e_w = (c_w, T_w, V_w)$: pre-faceoff context c_w (score state, zone start, period, rink, schedule, manpower, team-level priors), the teammate multiset T_w with co-presence seconds, and the opponent multiset V_w ;
- a lever vector $\ell_w \in \mathbb{R}^{17}$ of coach-controlled deployment quantities;
- an outcome vector $Y_w \in \mathbb{R}^{10}$: on-ice expected goals for/against (xGF, xGA), goals, shots, and four microstatistics, all attributed over s_w .

Windows aggregate to player-games and player-seasons in the obvious exposure-weighted way. We write Y^{60} for per-60-minute rates and $\text{xGF}\% = \text{xGF}^{60} / (\text{xGF}^{60} + \text{xGA}^{60})$.

2.2 The estimand

Fix a player p with history $\mathcal{H}_p(t_0)$ up to a freeze date t_0 , and a *hypothetical* environment distribution $\mathcal{E}'$ — in the product, the empirical window distribution of a destination team’s line slot, with teammates optionally pinned by the user. The target is the interventional mean over a horizon H (first 40 games):

$$\theta(p, \mathcal{E}', t_0) = \mathbb{E}[\bar{Y}_{p,H}^{60} \mid \text{do}(E \sim \mathcal{E}'), \mathcal{H}_p(t_0)], \quad (1)$$

reported as $\text{xGF}\%$ with an interval $[\theta_L, \theta_U]$ whose nominal coverage must match realized coverage on historical interventions (Section 8.3).

2.3 Identification

Equation (1) is identified from observational data under the following assumptions, each of which is engineered toward rather than assumed:

Assumption 1 (Overlap on the intervention class). $\mathcal{E}'$ lies within the support of environments realized by players comparable to p . Enforcement: the intervention class is restricted to environment swaps — of which the ledger contains 729 natural instances — and a support gate refuses queries outside observed ranges (Section 8.2).

Assumption 2 (Conditional ignorability given talent and context). $Y_w \perp E_w \mid (z_p, c_w, \ell_w)$, where z_p is player ability. Deployment is assigned by coaches on their beliefs about ability; conditioning on an explicit, externally estimated ability quantity z_p is what renders the assignment ignorable. Enforcement: z_p is estimated out-of-fold from a baseline that never observes player identity (Section 5) and enters the model through a dedicated channel (C2), so environment coefficients cannot absorb it.

Assumption 3 (Consistency / no interference across the swap). *The player’s ability transports across environments over the projection horizon; general equilibrium effects of the acquisition on the rest of the league are negligible at window scale.*

Remark 1. *Assumption 2 is the load-bearing one and it is only as good as z_p . The residual confounding from imperfect z_p is not argued away: it is priced — the conformal layer is fit on realized intervention errors and therefore absorbs, empirically, whatever ignorability violations survive (Section 10, T1–T3).*

3 Relation to Prior Approaches

Marcel-class projections (weighted multi-season averages with regression to the mean and an age adjustment) are the de facto industry standard; they are environment-blind by construction and serve as our primary baseline. **RAPM/WAR-family models** estimate context-adjusted player value by ridge regression over on-ice indicator matrices; they share our goal of separating player from context but produce retrospective value, not forward projections under specified interventions, and are afflicted by the collinearity of persistent line combinations (our T3). **Deep sequence models** of play-by-play data optimize predictive likelihood and inherit the deployment confound wholesale. Within the causal-ML taxonomy, our system is closest to an S-learner with an explicitly engineered adjustment set, wrapped in split-conformal inference; the novelty we claim is not any single component but the enforcement architecture binding them (C1–C4), which we have not seen deployed jointly in a production sports system.

4 Data Substrate and Forensic Repairs

4.1 Perfect Windows: a contracted training surface

All modeling reads from a single table, `perfect_windows_v2` (6,490,872 rows; seasons 2018-19 through 2024-25; 5v5), assembled by joining audited sources and validated at build time against five machine-checked contracts:

1. every player-performance feature is lagged and empirical-Bayes shrunk, and ships with its effective sample size n_{eff} ;
2. every context feature is measurable at the window’s opening faceoff (PRE-timing);
3. people lists (`with_ids`, `vs_ids`) are fixed length $K=5$, sorted by co-presence seconds descending (a legacy system silently consumed *unsorted* lists as if they were top- K ; the raw sources are confirmed unsorted);
4. baseline predictions are strictly out-of-fold, with fold provenance retained;
5. outcome columns are targets and are refused as features by the validator.

4.2 Forensic findings

Three corruptions were found by the validation gate before any training run, in data that had already been used by predecessor systems:

1. **Null outcomes.** The player-game observation file underlying all residual computation carried $y \equiv 0$ in every one of 761,823 rows — a silently failed join. Every “residual” was $-\mu$. This retroactively explains the predecessor uncertainty model’s 12% empirical coverage at a nominal 80% (its $\bar{z} = -1.87$ is exactly the missing-actuals artifact). Outcomes were rebuilt from per-game ground truth with a 100.0% join rate across seven seasons.

2. **Duplicate exposure.** The 10.29M-row window spine contained 1,491,039 full-row duplicates spanning all eight seasons (a combine-step defect), double-counting exposure downstream; deduplication left zero key conflicts. The corruption had propagated into the out-of-fold prediction files (12.9% duplicate keys), which are now key-deduped at every load.
3. **Global baseline miscalibration.** The environment baseline underpredicts on-ice xG by a season-dependent factor (mean residual +0.9 to +1.9 per 60). Rather than rescale the baseline (a label-aware operation with leakage risk), talent is defined *league-relatively* by per-season demeaning (Section 5), rendering the estimator invariant to any global miscalibration.

5 The Talent Prior

5.1 Construction

Let r_{pg} denote player p 's game- g on-ice xGF^{60} residual against the out-of-fold, identity-blind environment baseline, demeaned within season $s(g)$ by the TOI-weighted league mean:

$$\tilde{r}_{pg} = r_{pg} - \frac{\sum_{q,h: s(h)=s(g)} w_{qh} r_{qh}}{\sum w_{qh}}, \quad w_{pg} = \text{TOI}_{pg} \text{ (hours)}. \quad (2)$$

Talent state ahead of any date t is an exponentially decayed, *lagged* running moment set with half-life λ (default 40 games):

$$m_p(t) = \frac{\sum_{g:t_g < t} \rho^{\Delta(g,t)} w_{pg} \tilde{r}_{pg}}{\sum_{g:t_g < t} \rho^{\Delta(g,t)} w_{pg}}, \quad n_p(t) = \sum_{g:t_g < t} \rho^{\Delta(g,t)} w_{pg}, \quad \rho = 2^{-1/\lambda}, \quad (3)$$

implemented so the state recorded at each game excludes that game (verified by unit test: the running-statistic recorder emits before ingesting). The served quantity is empirical-Bayes shrunk toward the league (zero, post-demeaning):

$$z_p(t) = \frac{n_p(t)}{n_p(t) + K} m_p(t), \quad \hat{\text{se}}_p(t) = \max\left(\sqrt{v_p(t)/n_p(t)}, \text{se}_{\min}\right), \quad (4)$$

with K selected by out-of-time predictive risk over a grid (never hand-set) and a hard floor $\text{se}_{\min}$ replacing a legacy pathology in which 89.6% of low- n rows carried a collapsed, identically zero standard error. Defensive talent uses $-\text{xGA}$ residuals with the same machinery.

5.2 Face validity

On the first correctly computed residuals in the project's history, the untuned prior ranks McDavid first, Draisaitl second, and MacKinnon third among players with $n_{\text{eff}} > 15$ TOI-hours, with Kucherov, Crosby (age 37), Makar, and Barzal in the top fifteen and Kucherov's *defensive* component negative. We treat this as a necessary sanity property, not a result; the results are Section 9.

6 The Environment Tower

6.1 Design principle: the focal player has no embedding

The predecessor encoder assigned every player a learned identity vector and then fought a counter-intelligence war (gradient-reversal scrubbers, stream firewalls) to keep that vector from memorizing

deployment and team. We remove the disease instead of treating it: the focal player enters the network only through $(z_p, \widehat{se}_p, n_p)$ and explicit bio features (age, age², handedness, position). There is no lookup keyed on his identity, hence nothing to scrub. Teammates and opponents *do* receive embeddings — from the focal player’s vantage point, other players are environment, and environment is precisely what the intervention swaps.

6.2 Architecture

Inputs are restricted to a frozen legal schema (Section 6.3). Categorical context c_w maps through embeddings; continuous team priors pass through a normalized linear block. The people payload is encoded by seconds-biased attention pooling: for teammate embeddings $u_1, \dots, u_K$ with co-presence seconds $s_1 \geq \dots \geq s_K$,

$$\alpha_k = \text{softmax}_k(u_k^\top q + \log \max(s_k, 1)) \cdot \mathbb{I}\{s_k > 0\}, \quad h_{\text{with}} = \sum_k \alpha_k u_k, \quad (5)$$

and symmetrically for opponents. The tower output is $h_{\text{env}} = f_\theta(c_w, h_{\text{with}}, h_{\text{vs}}) \in \mathbb{R}^{256}$. Two training-time corruptions are integral to the design: *teammate dropout* (random masking of WITH identities) prevents h_{env} from fingerprinting the focal player through his habitual linemates and simultaneously matches the serving regime, where lineups are partially hypothesized; unseen identities at serving time resolve to pooled position×role fallback vectors, with degradation under synthetic unknowns measured as an explicit test.

6.3 The legal input schema

Every candidate feature was classified by a single criterion — *can the scenario builder reproduce it for a hypothetical destination from pre- t_0 information?* — into SYNTH (exactly constructible), MARG (served as the destination slot’s empirical distribution via bootstrap), or BANNED. Notable rulings, each traced to the generating code rather than inferred from names:

- **matchup_quality_pct** — BANNED. Traced construction: opponents’ *full-game* TOI percentile within role (includes post-window deployment, i.e. coach reactions to the game’s own outcomes), weighted by *realized within-window* head-to-head overlap seconds. Matchup difficulty reaches the tower legally through opponent identities and lagged opponent priors instead.
- **standing_prior** — BANNED. One code path selects standings rows with `after_pk ≤ current gamePk`, admitting the game’s own result.
- **ai_OZ_start**, **ai_DZ_start** — ALLOWED. “ai” denotes *after icing*: an icing-forced zone start, fully determined at the window’s first second, and incidentally the cleanest deployment natural experiment hockey offers.
- All in-window shot-style counts and sums — BANNED as outcomes wearing feature costumes, notwithstanding their (real) predictive value: the scenario builder can never supply them, so any accuracy they contribute is train/serve fiction.

The rulings compile to an `assert_legal()` guard that refuses banned features at trainer construction, citing the trace in the raised error.

6.4 Acceptance probes

A checkpoint ships only if a five-probe scorecard passes, all probes fit on held-out splits (the dry-run on random embeddings exposed in-sample probe overfitting — $R^2 = 0.095$ on pure noise — which would have flunked clean encoders; probes were repaired before first use and validated to refuse a random encoder, catch a synthetically leaky one, and pass a synthetically clean one):

Probe	Readout from	Requirement	Rationale
P1	$h_{\text{env}} \rightarrow \text{team identity}$	accuracy $< 3 \times$ chance	env must not memorize employer
P2	$h_{\text{env}} \rightarrow \text{focal OZ share}$	$R^2 < 0.10$	env must not encode focal deployment
P3	$h_{\text{env}}/\text{talent} \rightarrow \text{next-season ability}$	$R^2 > 0.30$	the system retains real signal
P4	cross-season retrieval	top-5 > 0.5	representations stable across years
P5	twin-distance ratio	< 1.5	player moves must not fracture identity

7 The CIN Core

7.1 Trunk

With $x = [h_{\text{env}}; \phi_z(z_p, \widehat{\text{se}}_p, n_p); \phi_b(\text{bio})]$ projected to width $d=768$, the trunk applies $L=4$ pre-norm residual blocks with SwiGLU activation, each modulated by the lever vector through feature-wise linear modulation (FiLM):

$$\text{Block}(h) = h + \text{Drop}\left(\left(1 + \tanh W_g \ell\right) \odot \text{SwiGLU}(\text{LN}(h)) + W_b \ell\right). \quad (6)$$

FiLM gives levers clean intervention semantics — knobs modulate the computation rather than mixing additively with state — and the talent channel ϕ_z is a separate pathway whose contribution the trunk cannot re-derive from deployment inputs (C2 made structural).

7.2 Heads and objective

Ten heads emit $(\mu_j, \log \sigma_j)$ per window. The per-head loss is heteroscedastic Gaussian NLL for the semi-continuous xG rates and Poisson-type likelihoods for count heads (goals, shots, microstatistics), exposure-offset by $\log(s_w/60)$; 68.8% of windows carry $y_{\text{xGF}} = 0$, and a homoscedastic Gaussian on that margin is misspecified. Cross-task weights are learned (Kendall–Gal):

$$\mathcal{L} = \sum_{j=1}^{10} e^{-\eta_j} \overline{\mathcal{L}}_j^{(\omega)} + \frac{1}{2} \eta_j, \quad \omega_w = \underbrace{s_w}_{\text{exposure}} \cdot \underbrace{\left(1 + (\kappa - 1) \mathbb{I}\{g_w \leq 10\}\right)}_{\text{switch weight (C3), } \kappa=3}, \quad (7)$$

where g_w counts games since the focal player’s last team change: the loss concentrates on post-transition windows, the regime the product queries. Internal σ_j are used for training and uncertainty *ranking* only; they are contractually barred from human-facing intervals (Section 8.3). Training uses era-split checkpoints (strictly-before 2021/22/23/24) so that every backtested projection is freeze-legal, and a gradient-boosted twin trained on the identical legal features serves as a referee: the neural core does not ship unless it beats both the twin and the v0 estimator under the evaluation protocol of Section 9.

8 Serving

8.1 Scenario construction and the v0 estimator

Given $(p, t_0, \text{destination}, \text{line})$, the builder identifies the slot incumbent by pre- t_0 TOI rank, bootstrap-resamples $N=2000$ of the incumbent’s real pre- t_0 windows (preserving the joint distribution of context — score states, zone starts, schedule — rather than sampling marginals independently),

overwrites the traveling-player block with p 's frozen talent and bio, and pins user-specified linemates into the WITH slots. The minimal estimator is then

$$\widehat{\text{xGF}}^{60} = \mathbb{E}_{\hat{\varepsilon}'}[\mu_{\text{xGF}}^{\text{OOF}}] + z_p^{\text{off}}(t_0), \quad \widehat{\text{xGA}}^{60} = \mathbb{E}_{\hat{\varepsilon}'}[\mu_{\text{xGA}}^{\text{OOF}}] - z_p^{\text{def}}(t_0), \quad (8)$$

i.e. destination environment expectation plus transported talent — explainable in one sentence and, per Section 9, already sufficient to clear the industry baseline. The neural path (v1) replaces the first terms with trunk forward passes over the same synthetic windows, and is what makes linemate-level swaps and Tier-2 lever queries answerable at all: estimator (8) is structurally blind to linemate identity.

8.2 Support gating

Queries are tiered. Tier 1 (environment swap): answered, subject to slot support (≥ 200 real windows) and prior quality checks. Tier 2 (lever nudge): answered only within the focal archetype's observed lever range $[q_{05}, q_{95}]$ harvested from the validated lever-sweep corpus. Tier 3: the engine returns a refusal object, `REFUSED_OOS`, never a number. Refusal-as-output is, in our view, the single clearest behavioral difference between causal inference as engineering and as branding.

8.3 Conformal calibration

Intervals come from split-conformal inference on the *intervention population*: signed errors of frozen-at- t_0 projections on the historical ledger, binned by talent n_{eff} , with the standard exchangeability-based marginal coverage guarantee within bins. Every emitted interval carries its bin's *achieved* coverage; emission without it is a contract violation. Current global figures: nominal 80%, achieved 79.8%. The predecessor system's internal-variance intervals achieved 12% at the same nominal level — the difference between modeled and measured uncertainty is the credibility of the entire product.

9 Evaluation

9.1 Protocol

The ledger comprises every player–team transition 2021–2025 detected in our own data (teamId change across consecutive appearances), qualified at ≥ 1000 career 5v5 minutes pre-move and ≥ 20 GP post-move: 729 moves (196 in-season trades, 533 off-season). For each move, the world is frozen at t_0 : talent snapshot, destination templates, and baselines all read strictly pre- t_0 data (era-split models for any learned component). The target is realized on-ice xGF% over the first 40 post-move games. An executable leakage audit re-runs sampled moves with post- t_0 rows physically deleted and asserts bit-identical projections. Baselines: last-season carryover; Marcel (5/4/3 season weights, regression to league mean by minutes, age adjustment); and Marcel plus half the team-quality delta (the strawman a hostile reviewer proposes).

9.2 Results

Projector	RMSE (xGF%, first 40 GP)	Δ vs. Marcel	95% CI on Δ
Last-season carryover	7.47	+1.67	—
Marcel	5.80	—	—
CIN v0 (this work)	5.30	−0.50 (−8.6%)	[−0.67, −0.33]

$n = 635$ scoreable moves; the confidence interval is a 10,000-resample bootstrap of the paired RMSE difference and excludes zero by a wide margin. Sliced by move type, v0 attains 5.14 on in-season trades versus 5.37 on off-season moves — the system is *strongest* on the natural experiments closest to its identification story, and to the customer’s pain. Emitted 80% intervals achieve 79.8% coverage; interval half-widths ($\approx \pm 6.6$ points) are wide and honest, dominated by the sampling noise floor of a 40-game on-ice share ($\approx 3\text{--}4$ points), which no projector can beat and which we disclose rather than obscure. The v0 result is a floor in a second sense: the estimator uses one scalar per player per side and an environment *average*; the neural core exists to convert linemate identity and lever structure into margin beyond -8.6% , and ships only if it does (Section 7).

10 Threat Model

The complete 18-item audit (deployment confounding; talent–residual circularity; linemate collinearity; ledger survivorship at the ≥ 20 -GP filter, a collider whose bias applies symmetrically to all compared projectors; acquisition selection-on-fit; injury unobservables; role-assignment error; regression to the mean; era drift and short seasons; goaltender contamination of defensive signal; score-state effects; out-of-support extrapolation; the sampling noise floor; the three data corruptions of Section 4; baseline integrity; deadline-context shift; cold-start prior degeneracy; and uncertainty miscalibration) is maintained in the repository README with a disposition for each: SOLVED structurally, PRICED into conformal intervals, GATED at the API, or OPEN with scheduled work. The methodological stance is uniform: no threat is argued away; each is closed, measured, or refused.

11 Limitations and Ongoing Work

(i) The scalar talent prior cannot express *fit* — complementarities between player styles; v1 adds a low-dimensional, probe-gated style vector on the talent side, computed from pre- t_0 windows only. (ii) The talent–outcome loop is one-step EM; iterating with talent-aware refits is scheduled once the v1 harness exists. (iii) Shrinkage constant K selected at the grid boundary under a per-game criterion; refit against next-season aggregates is queued. (iv) Sensitivity of the headline to the survivorship filter (≥ 10 GP variant) and to residual duplicate effects in legacy μ aggregation are scheduled re-runs. (v) The league-translation layer for AHL/ECHL projection — the commercial extension — is an exercise in data acquisition on this architecture, not new methodology: the query class is identical.

Reproducibility

The system is a single contracted repository (validation gate, executable schemas, era-split training protocol, freeze-audited backtest, hash-sealed forward predictions at `seal/`); every number in this report is regenerated by numbered pipeline scripts 00–06 from raw sources, and forward projections for the 2026-27 season are frozen and SHA-256-sealed before puck drop, to be scored publicly.